

Project Planning Template

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Epic (Functional Req.)	Story No.	User Story / Task	Points	Priority	Members	Start	End
Sprint-1	Environment Setup	USN-1	As a developer, I can install all required Python packages (numpy, pandas, matplotlib, seaborn, scikit-learn, Flask) using pip.	2	High	All	Week 1	Week 1
Sprint-1	Environment Setup	USN-2	As a developer, I can create the complete project folder structure with all directories and placeholder files.	1	High	Vinay Mohan	Week 1	Week 1
Sprint-2	Import Libraries	USN-3	As a developer, I can import all required libraries and configure the Jupyter notebook environment with %matplotlib inline.	1	High	All	Week 1	Week 1
Sprint-2	Dataset Loading	USN-4	As a data analyst, I can load the HDI.csv dataset into a Pandas DataFrame and inspect its structure using .head() and .shape.	2	High	Navya	Week 2	Week 2
Sprint-2	Data Visualization	USN-5	As a researcher, I can view strip plots of Life Expectancy vs HDI and Mean Schooling vs HDI for the first 20 rows.	2	High	Satya Srikala	Week 2	Week 2
Sprint-2	Data Visualization	USN-6	As a researcher, I can view a correlation heatmap showing relationships between HDI and all numeric features.	2	High	Venkatathri vendra Reddy	Week 2	Week 2

Sprint-3	Preprocessing	USN-7	As a developer, I can select independent variables (Country, Life Exp, Schooling, GNI, Internet) and the dependent variable (HDI Score).	2	High	Midathana Muktananda	Week 3	Week 3
Sprint-3	Preprocessing	USN-8	As a developer, I can check null values using X.isnull().sum() and fill them using column mean imputation.	2	High	Midathana Muktananda	Week 3	Week 3
Sprint-3	Preprocessing	USN-9	As a developer, I can apply LabelEncoder to the Country column to convert it to numeric values.	1	High	Venkatathriveendra Reddy	Week 3	Week 3
Sprint-4	Train/Test Split	USN-10	As a developer, I can split the dataset into 90% training (175 rows) and 10% testing (20 rows) using train_test_split.	1	High	Vinay Mohan	Week 4	Week 4
Sprint-4	Model Training	USN-11	As a developer, I can train a LinearRegression model on the training dataset and generate predictions on test data.	3	High	Vinay Mohan	Week 4	Week 4
Sprint-4	Model Evaluation	USN-12	As a researcher, I can view the R-squared score and Actual vs Predicted scatter plot to evaluate model performance.	2	High	Navya	Week 4	Week 4
Sprint-5	Model Saving	USN-13	As a developer, I can serialize the trained model to HDI.pkl using Pickle for deployment.	1	High	Midathana Muktananda	Week 5	Week 5
Sprint-5	Flask Backend	USN-14	As a user, I can access the Flask app at localhost:5000 with home page, prediction form, and result page.	5	High	All	Week 5	Week 6
Sprint-6	Web Interface	USN-15	As a user, I can select a country from the dropdown and enter HDI indicators to get a real-time HDI prediction.	5	High	All	Week 6	Week 6

Sprint-6	Result Display	USN-16	As a user, I can view my predicted HDI score and category (Very High/High/Medium/Low) on the results page.	3	High	Satya Srikala	Week 6	Week 6
Sprint-7	Testing	USN-17	As a developer, I can test all Flask routes and verify predictions match expected HDI categories.	3	Medium	All	Week 7	Week 7
Sprint-7	Documentation	USN-18	As a team, we can create complete project documentation including all 8 milestone deliverables.	3	High	All	Week 7	Week 8